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Insights on Shale Gas Production Prediction via Combining Multimodal Machine Learning with Deterministic Geological Modeling

Ye Li, State Key Laboratory of Petroleum Resources and Prospecting, China University of Petroleum Beijing, Beijing, China / College of Petroleum Engineering, China University of Petroleum Beijing, China; Gang Hui, State Key Laboratory of Petroleum Resources and Prospecting, China University of Petroleum Beijing, Beijing, China / Department of Chemical and Petroleum Engineering, University of Calgary, Calgary, AB, Canada; Zhaochen Zhang, CNPC Greatwall Drilling Company, Panjin, China; Muming Wang, Department of Chemical and Petroleum Engineering, University of Calgary, Calgary, AB, Canada; Jing Li, State Key Laboratory of Petroleum Resources and Prospecting, China University of Petroleum Beijing, Beijing, China / College of Petroleum Engineering, China University of Petroleum Beijing, China; Zhiyang Pi, State Key Laboratory of Petroleum Resources and Prospecting, China University of Petroleum Beijing, Beijing, China / College of Petroleum Engineering, China University of Petroleum Beijing, China; Chenqi Ge, State Key Laboratory of Petroleum Resources and Prospecting, China University of Petroleum Beijing, Beijing, China / College of Petroleum Engineering, China University of Petroleum Beijing, China; Penghu Bao, State Key Laboratory of Petroleum Resources and Prospecting, China University of Petroleum Beijing, Beijing, China / College of Petroleum Engineering, China University of Petroleum Beijing, China; Yujie Zhang, State Key Laboratory of Petroleum Resources and Prospecting, China University of Petroleum Beijing, Beijing, China / College of Petroleum Engineering, China University of Petroleum Beijing, China; Xing Yang, State Key Laboratory of Petroleum Resources and Prospecting, China University of Petroleum Beijing, Beijing, China / College of Petroleum Engineering, China University of Petroleum Beijing, China; Yujie Zhang, State Key Laboratory of Petroleum Resources and Prospecting, China University of Petroleum Beijing, Beijing, China / College of Petroleum Engineering, China University of Petroleum Beijing, China; Fei Gu, Research Institute of Petroleum Exploration and Development, Beijing, China

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Abstract

Traditional machine learning models for shale gas production prediction usually ignore spatial reservoir heterogeneity. This paper develops a multimodal machine learning (MML) method integrating a deterministic geological model with multi-source petrophysical data. The model combines a CNN for spatial feature extraction and an ANN for tabular data analysis, significantly improving prediction accuracy. It is found that the MML model achieves a superior coefficient of determination (R^2 = 0.849) for 12-month Duvernay shale gas productivity forecasting, significantly outperforming the standalone ANN model (R^2 = 0.719). Furthermore, the MML approach successfully explains production discrepancies between horizontal wells with similar average reservoir properties—a challenge traditional ANN models fail to address—by capturing lateral formation variability. Validation on three representative wells confirms high prediction accuracy, with conformity rates of 89.1%, 91.8%, and 92.6%, respectively. This method not only enhances production forecasting but also provides a practical tool for optimizing horizontal well placement, enabling more efficient development of shale gas reservoirs. The study highlights the potential of multimodal machine learning in unconventional resource exploitation, providing a data-driven approach to reservoir management.

Introduction

In recent years, the triumph of the U.S. shale gas revolution has brought about profound shifts in the competitive dynamics and trade structures of the global natural gas market. These changes have exerted a notable influence on the worldwide oil and gas supply landscape as well as geopolitical relationships across the globe (Gerlagh et al., 2024; Kirat, 2021; Middleton et al., 2017; Wang et al., 2014). As part of the unconventional energy sector in North America, Canada has become the second country in the world after the United States to successfully develop shale gas (Zou et al., 2018; Sun et al., 2019). Experts and scholars from around the world have carried out in-depth studies on the advancement of shale gas resources, including key technologies for shale gas development (Wang et al., 2023; Lu et al., 2012; Chen et al., 2021), shale gas resource evaluation (Deng et al., 2023; Hui et al., 2022; Wang et al., 2022), and factors controlling sweet spots in shale gas (Qiu and Zou, 2020; Hui et al., 2021; Fang et al., 2024). Nonetheless, when it comes to shale gas productivity, there exists a multitude of influencing factors alongside intricate patterns within vast volumes of data (Xiao et al., 2022; Yi et al., 2024). Using traditional research methods has certain subjectivity and limitations, leading to an ambiguous grasp of the key controlling factors that influence productivity, and resulting in inaccurate predictions and evaluations of productivity, and poor practical application results (Bao et al., 2025; Tong et al., 2024; Lawal et al., 2024).

Machine learning algorithms, such as regression algorithms, neural network methods, decision learning, Bayesian methods, etc., as methods for mining massive information data, can detect hidden relationships between various main control factors and discover the main control factors that affect shale gas production capacity (Sheth et al., 2022; Hui et al., 2023a; Uncuoglu et al., 2022; Awoleke and Lane, 2011). Specifically, linear regression is the simplest but most powerful machine learning algorithm, typically used to preliminarily clarify linear relationships between variables (Luo et al., 2022; Wang et al., 2025; Wang and Chen, 2019). Neural networks simulate complex relationships between input and output variables by mimicking the behavior of biological neural networks (Wu et al., 2023; Ha and Jeong, 2021; Genuer et al., 2017). Decision learning is based on ensemble methods, generating a set of decision trees and reporting the average output, that is, fitting the residuals between each step goal and the current prediction result for prediction (Obregon and Jung, 2022; Wang et al., 2019; Bakouregui et al., 2021; LeCun et al., 2015). The Bayesian method is to infer the possible probability combinations of the involved parameters based on the established results (Van et al., 2021; Tang et al., 2021). Among them, neural network methods have shown significant advantages in processing large amounts of data and accurately obtaining predictive models, and have achieved good application results in shale gas fracturing optimization, production prediction, decline analysis, and other aspects (Nehzati et al., 2025; Zhou et al., 2025; Omidkar et al., 2024; Saporetti et al., 2022). However, the traditional machine learning models mentioned above neglect the fusion of multimodal data such as geological images and engineering geological parameters, resulting in large prediction errors and unexpected engineering designs.

Given this, it is urgent to construct a multimodal fusion machine learning framework that integrates geological images and geological engineering parameters of multiple information to break through the performance bottleneck of single-modal models. This work proposes a multimodal machine-learning (MML) approach based on the deterministic geological model. This functional distinction allows the CNN component to excel at extracting intricate features from image-based data—such as spatial hierarchies and visual patterns—while the ANN module specializes in analyzing structured, row-and-column formatted data, making it well-suited for tasks involving numerical or categorical attributes. The input from both modalities was integrated and examined by a fusion module.

Field Background

The Duvernay shale gas reservoir near Fox Creek, Alberta, was selected as our target to forecast the gas production using the proposed MML approach (Fig.1). The Duvernay shale gas resource has long been regarded as a world-class unconventional resource (Wang et al., 2018; Hui et al., 2023b; Chen and Jiang, 2020). Duvernay shale gas production has been dramatically increased since 2010 due to hydraulic fracturing. The Duvernay shale covers approximately 2.42 × 10^4 km^2 and has discovered reserves of 251 × 10^8 t of crude oil, 115 × 10^8 t of liquid hydrocarbons, and 23 × 10^{12} m^3 of natural gas (Kleiner and Aniekwe, 2019; Leshchyshyn et al., 2016; Hui et al., 2024).

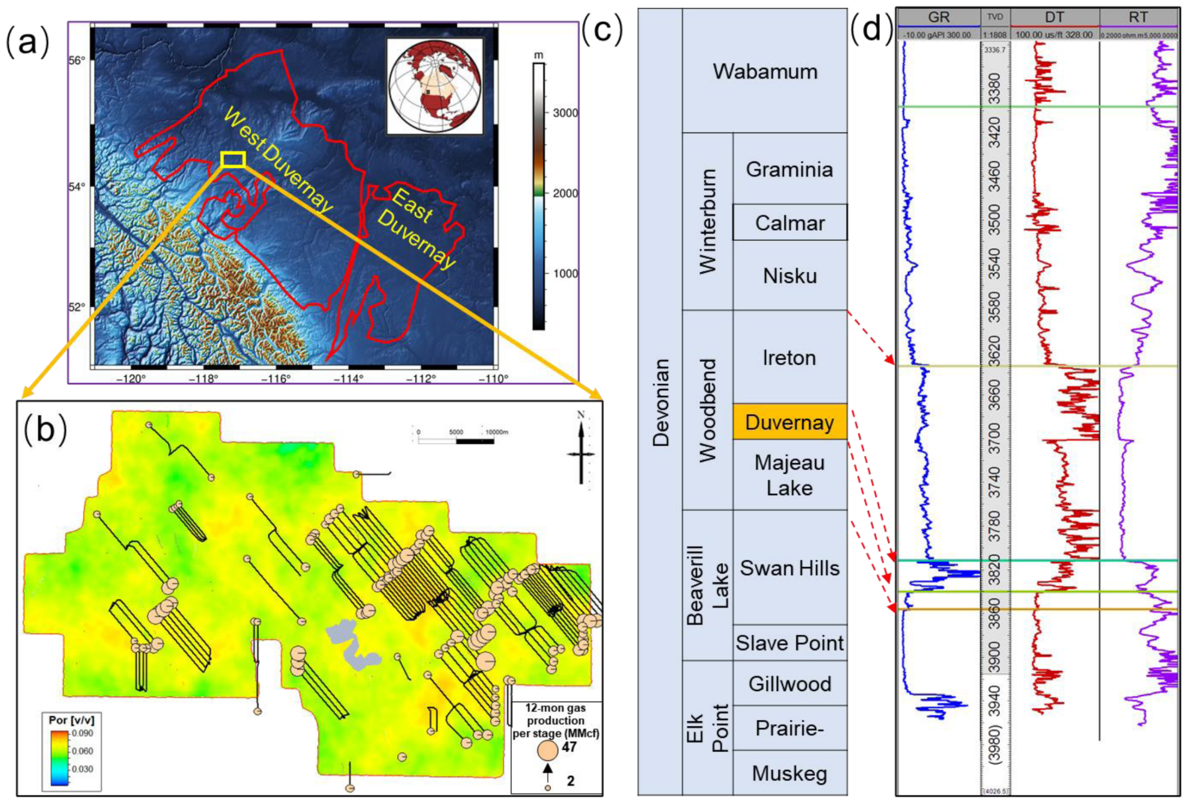

Figure 1—Geological details of the study area. (a) Position of the study area within the Western Duvernay Basin, Canada. (b) Plan view of the studied wells. The base map depicts formation porosity; black lines indicate horizontal well path, and pink circles represent the 12-month equivalent gas productivity. (c) Relevant formations in the study area, with the Duvernay Formation highlighted in orange. (d) Typical logging responses of the Duvernay shale, where GR stands for gamma ray, DT for acoustic logging, and RT for deep resistivity.

The study area is situated in the Western Duvernay Basin, Canada (Fig. 1a). Data pertaining to geology, completion operations, logging results, testing outcomes, and fracturing treatments of 133 horizontal wells were compiled from the Geoscout database (accessed on July 1, 2025, at https://www.geologic.com/products/geoscout) for this research. Experiments encompassing mineralogy, geophysics, geochemistry, and geomechanics were conducted using core samples obtained from 10 cored wells. The plan view of these studied wells is presented in Fig. 1b, where the base map displays formation porosity, black lines denote the horizontal well path, and pink circles stand for the 12-month equivalent gas productivity. The relevant formations in the study area and the characteristic logging responses of the Duvernay shale reservoir are illustrated in Figs. 1c and 1d, respectively. The Duvernay shale is distinguished by high gamma ray (GR) values, medium acoustic (DT) readings, and high resistivity (RT) measurements.

The studied shale formation exhibits a thickness ranging from 21 to 58 meters, reflecting lateral variation in depositional characteristics and diagenetic processes (Dunn and Humenjuk, 2014). The reservoir is characterized by inherently poor physical properties, typical of organic-rich shale systems (Figs. 2a-2e). Specifically, the average effective porosity is measured at 3.9%, indicating limited pore volume available for hydrocarbon storage (Hui et al., 2023c). In addition, the matrix permeability is extremely low, averaging only 130 nD, which significantly restricts fluid flow under natural conditions (Fig.2a). These characteristics highlight the necessity for advanced stimulation techniques, such as hydraulic fracturing, to enhance gas recovery and establish effective production pathways within the tight shale matrix.

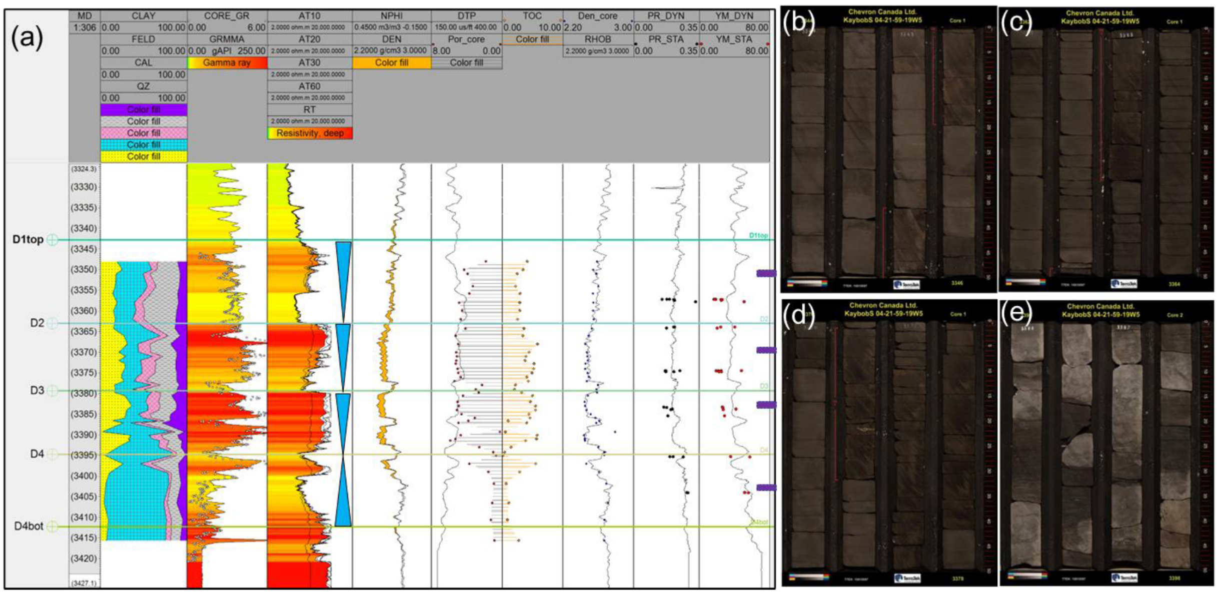

Figure 2—Petrophysical and geomechanical data concerning the Duvernay shale formation. (a) The reservoir properties of the key well (Hui et al., 2023c). In the first column. The purple boxes in the last column indicate the coring locations for (b) to (e). (b-e) Show the core observation results of the key well.

Statistical analysis of fracturing treatments for horizontal wells reveals that each horizontal well comprises 37 fracturing stages with a horizontal interval length of 2,116 meters on average. Additionally, the average quantities of proppant placed and fluid injected stand at 5,650 tons and 38,253 cubic meters, respectively, while the average fracture depth is 2,478 meters below the surface. Owing to fracturing stimulation, the initial 12-month shale gas equivalent for each stage—encompassing both dry gas and condensate gas—varies from 2 to 47 million cubic feet, with a mean value of 18.7 million cubic feet (MMcf).

In addition, a high-resolution 3D reflection seismic survey was used to determine the reservoir's lateral variability. This seismic dataset provides precise imaging of subsurface structures and stratigraphy, allowing for the extraction of planar reservoir parameters such as thickness, lithofacies distribution, and fracture development with greater spatial precision. Integrating seismic-derived features with well-level data enables a more informed and geologically consistent study of shale gas production (Fig.3).

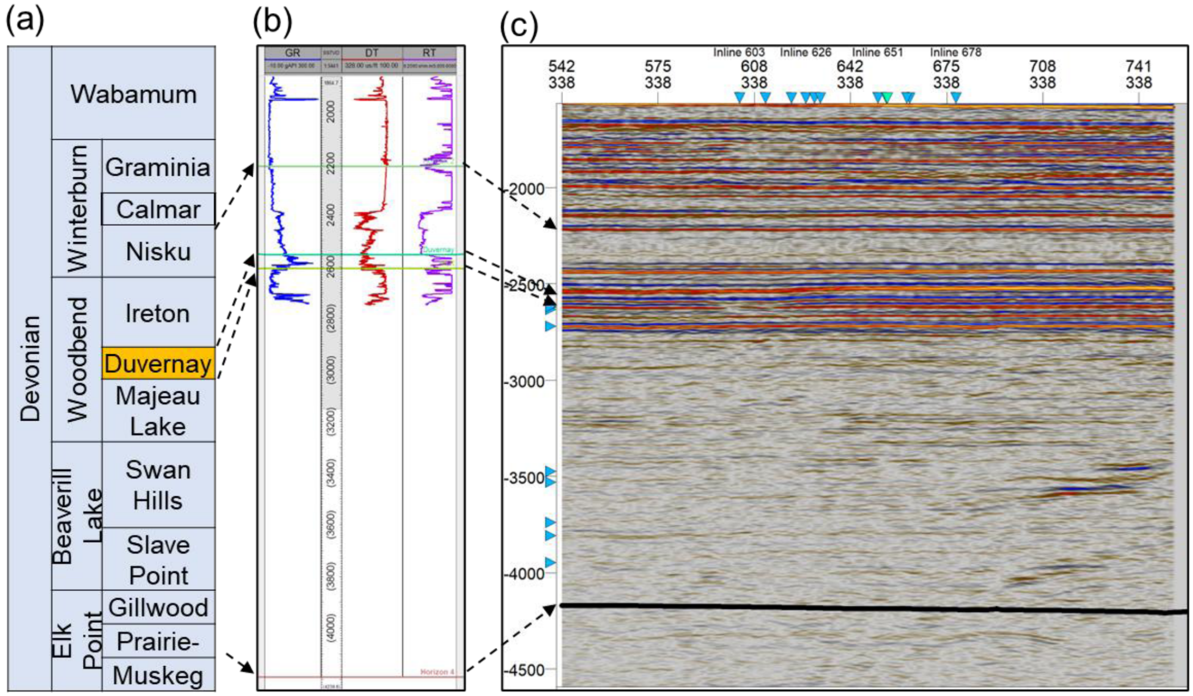

Figure 3—Well to seismic calibration. (a) Related formations in the studied region. (b) Logging response of different formations from a coring well. (c) Cross-section of the 3D seismic survey showing the features of associated formations.

Based on prior research, the stratigraphy in this area was specifically examined. The Cambrian and Precambrian basement formed beneath the Devonian sediments, which are where the Duvernay formation was deposited, as seen in Fig.3a. Then, using the well-logging characteristics of a straight coring well, characteristic logging responses in the Duvernay formation were identified (Fig. 3b). A 3D seismic survey is used to identify the Precambrian basement based on previous works, because there is no sonic log available in the basement depth (Fig. 3c).

Methodology

This paper presents a multimodal machine-learning (MML) approach to forecast the shale gas production. A deterministic geological model, constrained by well-logging and 3D seismic data, is combined with a multimodal machine learning method to forecast shale productivity.

A multimodal machine learning model is developed by integrating two distinct neural network architectures: a convolutional neural network (CNN) module for processing tabular datasets and an artificial neural network (ANN) module for analyzing visual datasets. This hybrid approach leverages both structured numerical data and spatial geological representations to enhance production forecasting accuracy.

To ensure robust geological inputs, a deterministic geological model is constructed using well logging interpretations and 3D seismic inversion data. This model delivers high-resolution subsurface characterization, capturing key reservoir properties like porosity, permeability, gas saturation, and shale thickness along the path of horizontal wells. These parameters are essential for understanding near-wellbore reservoir heterogeneity and its impact on production performance.

The CNN module is purposefully engineered to extract verisimilitude features from visual datasets, including seismic attribute maps and spatial distributions of reservoir properties. Meanwhile, the ANN module processes conventional tabular datasets, including well completion parameters, hydraulic fracturing data, and production history.

Finally, a fusion module combines and processes inputs from both modalities, integrating geological insights with engineering parameters to generate reliable shale gas production predictions. This multimodal framework addresses the limitations of traditional methods by incorporating both regional reservoir characteristics and well-specific completion data, leading to improved consistency and predictive performance.

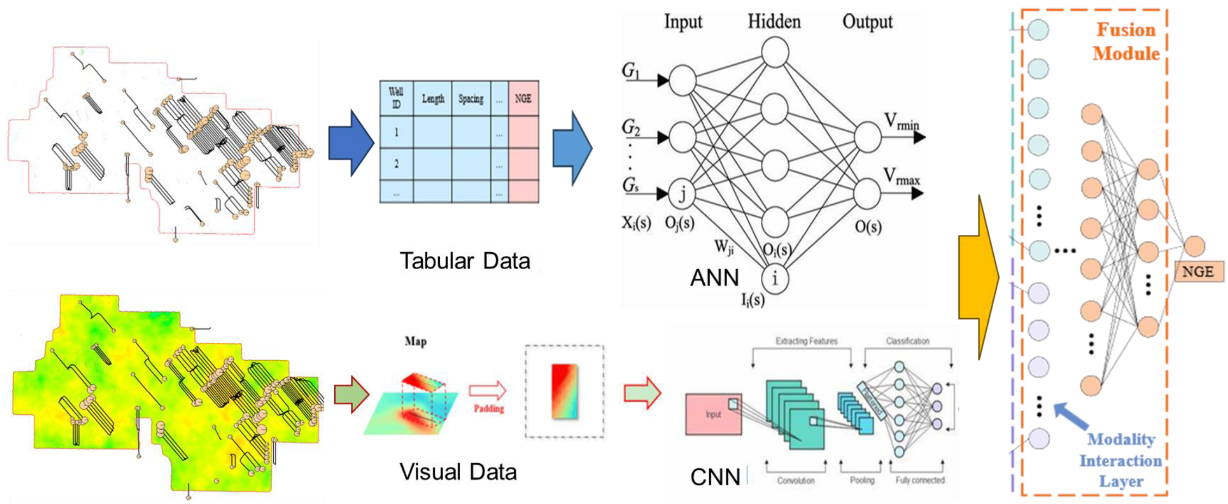

Figure 4—Workflow for the MML approach to forecast shale gas production. The CNN module is employed for processing tabular datasets, while the ANN module is utilized for analyzing visual datasets (Wang et al., 2025). A fusion module combines and processes inputs from both modalities, integrating geological insights with engineering parameters to generate reliable production predictions.

Based on the statistical results of the horizontal section of the horizontal well, the length of the horizontal well cross-section screenshot was selected as 3,500 m. At the same time, based on the results of the width sensitivity analysis, the width of the horizontal well cross-section screenshot was selected as 1,000 m (Wang et al., 2025). Screenshots of the six geological parameters of the input attributes (i.e., pressure, thickness, depth, porosity, permeability, and saturation) were taken for CNN image training, as shown in Figs. 5a-5c.

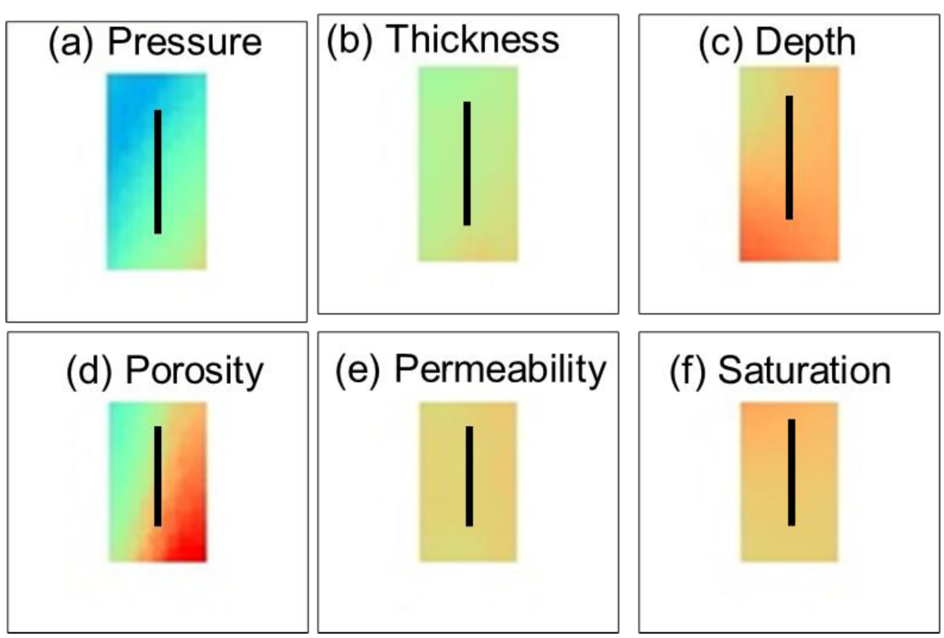

Figure 5—High-level information from the visual dataset. (a) Pressure. (b) Thickness. (c) Depth. (d) Porosity. (e) Permeability. (f) Saturation.

Results and Discussion

Input tabular data of geological and operational parameters

The deterministic geological model is constructed by integrating well logging interpretation results with reflection-based 3D seismic inversion data, enabling a more accurate spatial characterization of subsurface formations. This model provides a robust framework for delineating key reservoir features and heterogeneities. Comprehensive, high-resolution information regarding vital geological properties—including porosity, permeability, gas saturation, and shale thickness—is extracted with improved accuracy, particularly in the vicinity of horizontal well trajectories, where production performance is highly sensitive to local reservoir quality. Table 1 shows the statistical results of input and output geological and operational parameters.

Table 1—Statistics of input and output geological and operational parameters.

Type	Parameter	Unit	Min	Max	Avg
Output	12-month production	MMcf	2	47	18.7
Geological	Formation pressure	MPa	36.7	67.1	51.5
	Thickness	m	21	58	36.8
	Burial depth	m	2311	2644	2478
	Porosity	%	2.5	8.6	3.9
	Permeability	nD	0.1	1160	130
	Gas saturation	%	59.1	92.2	80.4
Operational	Horizontal length	m	180	3438	2116
	Number of stages		10	87	37
	Fracturing fluid	m3	18	43300	38253
	Proppant mass	t	24	14130	5650

Formation pressure. The data on formation pressure are sourced from the publicly available database (Eaton and Schultz, 2018). The isochore interpretation method is utilized to estimate the spatial distribution of formation pressure. Formation pressure ranged from 36.7~67.1 MPa, with the mean value of 51.5 MPa, with an increasing trend from east to west (Fig.6a).

Net thickness. The data on net thickness at the well site are sourced from the stratigraphic correlation results from well logging response of different straight shale wells (Hui et al., 2023d). The data of net thickness between wells are derived from the 3D seismic interpretations (Fig.3c), which are combined with the well site data to depict the spatial distribution of net thickness. Net thickness ranged from 21~58 m with an average value of 36.8 m (Fig.6b).

Buried depth. The data on buried depth are also sourced from the logging-based stratigraphic correlation (Pawley et al., 2018). Similar to the net thickness, the 3D seismic interpretations are also utilized to estimate the spatial distribution of buried depth. It is shown that the buried depth ranged from 2311~2644 m in vertical depth and averaged to be 2478m, displaying an increasing trend from SW to NE (Fig.6c).

Porosity. Porosity is sourced from core experiments, and the porosity model is constructed from the core-logging integration. Spatial distribution of porosity is constructed based on well-logging interpretation and 3D seismic inversions. Porosity in the studied region ranged from 2.5%~8.6% and averaged to be 3.9% (Fig.6d).

Permeability. Permeability is sourced from core experiments, and the model is also constructed via core-logging integration. Spatial distribution of permeability follows the pattern of porosity. It is shown that permeability in the examined region ranged from 0.1~1160 nD with an average value of 130 nD (Fig.6e).

Gas saturation. Saturation is also sourced from core experiments. Here, the gas saturation refers to the free gas saturation and the absorbed gas saturation is not considered in this work. Similar to porosity and permeability, the spatial distribution of saturation is also constructed via well-logging interpretation and 3D seismic inversions. The average value of free gas saturation in the examined region is 80.4% and has a range from 59.1~92.2% (Fig.6f).

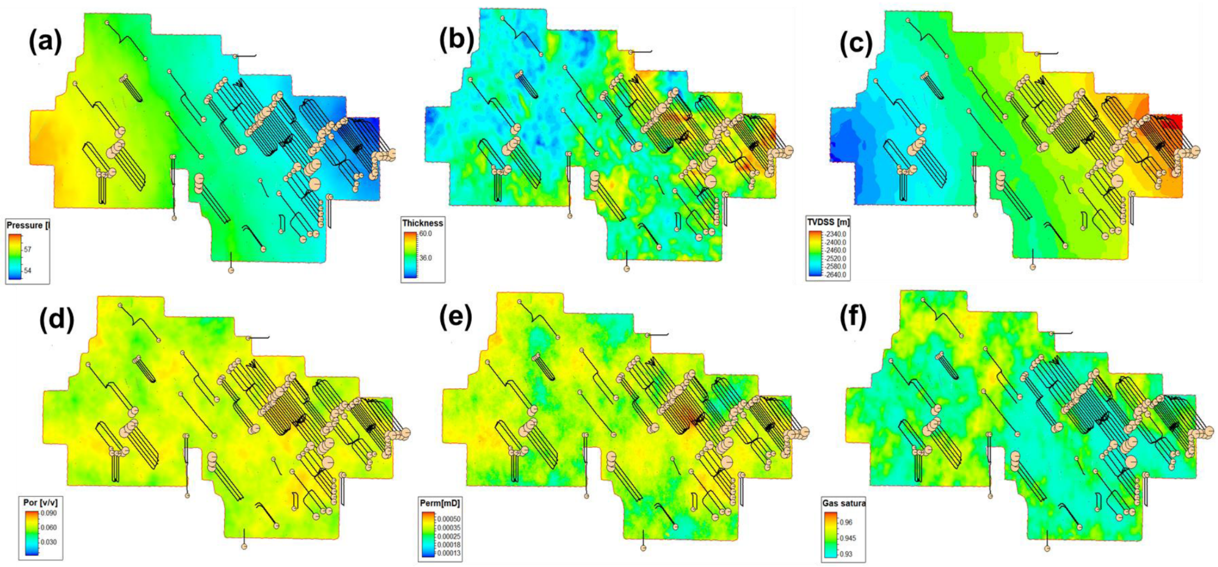

Figure 6—Map view of combined well-logging and 3D-seismic-restrained reservoir properties. (a) Formation pressure. (b) Shale thickness. (c) Buried depth. (d) Porosity. (e) Permeability. (f) Gas saturation.

Operational parameters. Operation parameters include the first 12-month gas production equivalence, horizontal length, number of stages, cumulative fluid injection, and cumulative proppant mass. Operational parameters are sourced from the GeoScout database (https://www.geologic.com/products/geoscout/). The average horizontal length and number of stages are 2116m and 37, respectively, which ranged from 180~3438 m and 10~87, respectively. Meanwhile, the average injection volume of fracturing fluid and proppant mass is 38253 m^3 and 5650 t, respectively, ranging from 18~43300 m^3 and 24~14130 t, respectively.

Input visual data of geological and operational parameters

The CNN module is purposefully engineered to extract verisimilitude features from visual datasets, such as seismic attribute maps, fracture network distributions, and reservoir property grids. By leveraging convolutional operations, this module effectively captures local patterns, geological discontinuities, and regional heterogeneity that influence shale gas productivity (Figs.7a-7g).

To synthesize insights from both data modalities, a fusion module dynamically integrates and processes inputs from the CNN (visual data) and ANN (tabular data) branches. This fusion mechanism employs feature concatenation, attention-based weighting, or cross-modality learning to ensure that geological characteristics and engineering parameters collectively inform the final prediction. The unified output provides a more robust and accurate forecast of shale gas production, accounting for both subsurface complexity and completion design effects.

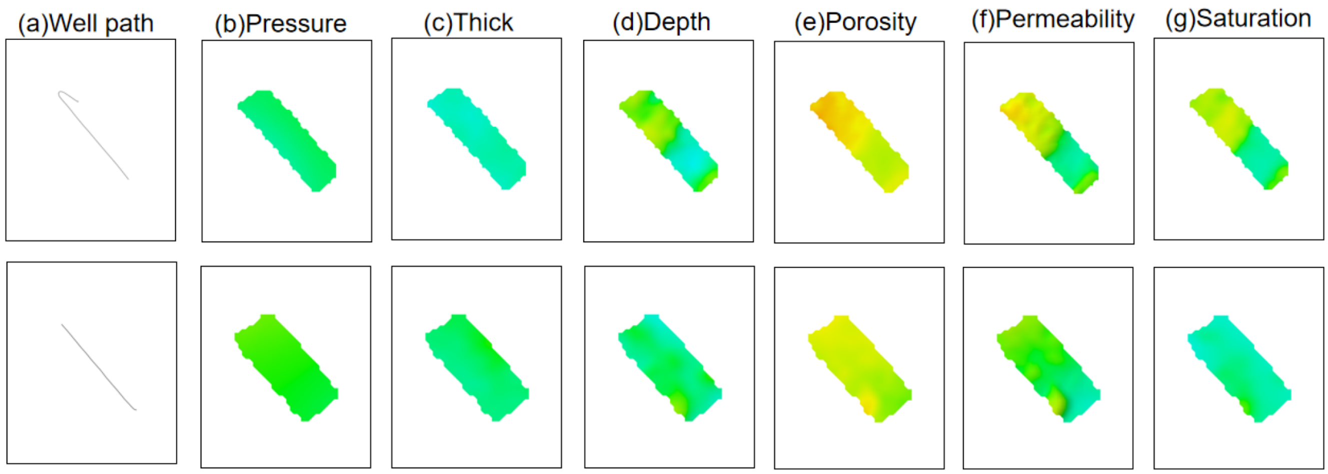

Figure 7—The well path and input visual data processing by CNN. (a) Well path (b) Formation pressure. (c) Shale thickness. (d) Buried depth. (e) Porosity. (f) Permeability. (g) Gas saturation.

Prediction results of the multimodal approach

To evaluate the predictive performance of different machine learning approaches, a comparative analysis was conducted involving a multi-modal model (MM) and three commonly used baseline algorithms: artificial neural networks (ANN), extreme gradient boosting (XGBoost), and support vector machines (SVM). Each model was independently trained and tested over 50 runs to account for random initialization and variability, and the coefficient of determination (R^2) from each run was visualized using box plots.

As illustrated in Fig. 8, the vertical axis represents the R^2 values, while the blue and red box plots correspond to the training and testing datasets, respectively. The MM model consistently achieves the highest median R^2 values on both training and testing sets, with a median of 0.849, outperforming the baseline models—ANN (0.719), XGBoost (0.740), and SVM (0.670). It can be seen that the MM model exhibits good fitting to the training data and also achieves effective generalization to unseen data.

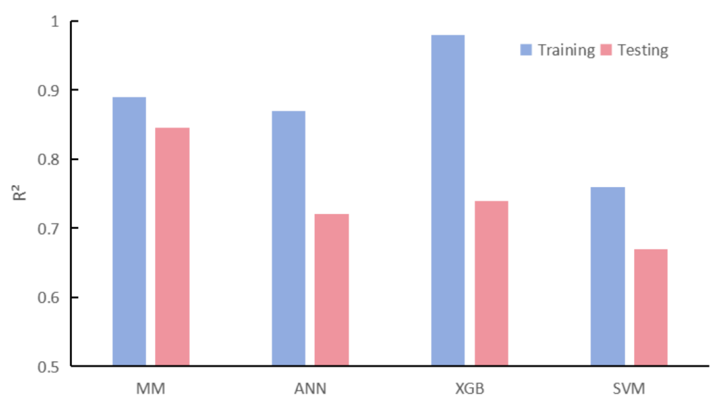

Figure 8—Prediction results of multi-modal model (MM), artificial neural networks (ANN), extreme gradient boosting (XGBoost), and support vector machines (SVM).

All models exhibit relatively high R^2 values on the training set, suggesting adequate fitting capacity. However, their generalization performance on the testing set differs notably. XGBoost and SVM show a significant drop in testing R^2 along with increased variability, indicative of overfitting.

In conclusion, while all models exhibit reasonable fitting ability, the multi-modal model achieves superior and more consistent performance across both datasets, highlighting its effectiveness in integrating heterogeneous input features.

Application of a multimodal approach

To demonstrate the application of the proposed multimodal modeling approach, three representative wells (i.e., Well A, Well B, and Well C) are selected based on their high cumulative gas production over 12 months (Table 2). These wells were chosen to reflect distinct reservoir conditions, specifically low, medium, and high porosity scenarios, thereby capturing a broad spectrum of geological heterogeneity.

Table 2—Statistics of input and output parameters.

Well Index	Porosity	Mean/Std	Tab	Pressure	Mean/Std	Tab	Thickness	Mean/Std	Tab	Actual Production (MMcf)	Predicted Production (MMcf)	Conformity rate (%)
A		0.04/0.01	0.03		38.67/0.009	37.6		47.7/1.1	57.0	9.2	10.2	89.1
B		0.06/0.02	0.06		40.82/0.004	40.8		57.5/1.7	60.1	13.4	12.3	91.8
C		0.08/0.02	0.11		44.66/0.001	46.1		60.1/4.3	49.0	17.6	18.9	92.6

For each representative well, key parameters derived from the visual data—such as seismic attributes and inversion-based property maps—were extracted and analyzed. The corresponding statistical descriptors, including the mean and standard deviation (std), were computed to quantify spatial variability and inform feature distribution. In parallel, relevant tabular (Tab) data, such as well logs and engineering parameters, were compiled to provide complementary numerical inputs. The integration of these multimodal features enables a comprehensive evaluation of reservoir behavior and supports accurate production prediction across varying porosity regimes.

The proposed method was successfully applied to the intelligent production prediction of shale gas wells in the Kaybob area. Based on the MM model proposed in this paper, the predicted 12-month shale gas production of Well A, Well B, and Well C reached 10.2, 12.3, and 18.9 MMcf, respectively, corresponding to the actual production of 9.2, 13.4, and 17.6 MMcf, respectively. Consequently, the rate of conformity for three wells is 89.1%, 91.8% and 92.6%, respectively. This work demonstrates a novel and reliable multimodal fusion framework for hydrocarbon production forecasting, offering a new technical pathway and methodological advancement for intelligent modeling in unconventional reservoirs.

Conclusions

This paper develops a multimodal machine learning (MML) method integrating a deterministic geological model with multi-source petrophysical data. The model combines a CNN for spatial feature extraction and an ANN for tabular data analysis, significantly improving prediction accuracy. The conclusions are drawn below:

1. The deterministic geological model is constructed by integrating well logging interpretation results with reflection-based 3D seismic inversion data, enabling a more accurate spatial characterization of subsurface formations.
2. The MML model achieves a superior coefficient of determination ($R^2 = 0.849$) for 12-month Duvernay shale gas productivity forecasting, significantly outperforming the standalone ANN model ($R^2 =$

0.719). Furthermore, the MML approach successfully explains production discrepancies between horizontal wells with similar average reservoir properties—a challenge traditional ANN models fail to address—by capturing lateral formation variability.
3. Validation on three representative wells confirms high prediction accuracy, with conformity rates of 89.1%, 91.8%, and 92.6%, respectively. The study highlights the potential of multimodal machine learning in unconventional resource exploitation, providing a data-driven approach to reservoir management.

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